

UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Artificial Intelligence Laboratory
2	Course Code	CSE 3812
3	Trimester and Year	Spring 26
4	Pre-requisites	CSE 2218: Data Structure and Algorithms II Laboratory
5	Credit Hours	1
6	Section	C
7	Class Hours	Saturday: 2:00 PM -04:30 PM
8	Classrooms	727
9	Faculty Name	Md. Tanvir Raihan (MTR)
10	Email	tanvir@cse.uiu.ac.bd
11	Office	419 (C)
12	Counselling Hours	Check ELMS
13	Text Book	 Artificial Intelligence - A Modern Approach (3rd Edition).pdf
14	Reference	None
15	Course Contents (approved by UGC)	Basics of data science with Python programming, Informed search strategy, Local search and variants, Dataset preprocessing, Supervised learning algorithm, Unsupervised learning algorithm, Concept of training and validation

16	Course Outcomes (COs) and Mapping																	
CO	Statement	Bloom's Domain	Program Outcome	Knowledge Profile	Complex Problem	Engineer-ing Activities												
CO1	Demonstrate application of modern tools for data preprocessing and analysis.	P	PO5 Modern Tool Usage	Engineering fundamentals (K3) Specialist knowledge (K4) Engineering practice (K6)	Depth of Knowledge (P1) Conflicting requirements (P2)	-												
CO2	Understand and apply search strategies for solutions of problems with complete information.	C	PO2 Problem Analysis															
CO3	Understand and apply reasoning strategies for inference in the presence of incomplete information.	C	PO2 Problem Analysis															
CO4	Understand and develop intelligent algorithm capable of learning from data	C	PO1 Engineering Knowledge															
17	Teaching Methods	Lecture, demonstration																
18	CO with Assessment Methods	<table><tr><td>CO</td><td>Assessment Method</td><td>(%)</td></tr><tr><td>-</td><td>Attendance</td><td>5%</td></tr><tr><td>CO1, CO2, CO3, CO4</td><td>Offline, Assignment and Mid Exam, Final Exam</td><td></td></tr><tr><td></td><td></td><td></td></tr></table>					CO	Assessment Method	(%)	-	Attendance	5%	CO1, CO2, CO3, CO4	Offline, Assignment and Mid Exam, Final Exam				
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-	Attendance	5%																
CO1, CO2, CO3, CO4	Offline, Assignment and Mid Exam, Final Exam																	

19	Lab Outline		
Class	Topics	COs	Activities
1	Python Tutorial I	1	Lecture
2	Python Tutorial II	1	Lecture
3	Class Test (Python)	2	Lecture
4	A* Search Theory and Assignment declaration	-	Evaluation
5	Assignment 1 evaluation, CSP theory	3	Lecture
6	Mid Exam	-	Evaluation
Theory Mid Terms			
7	Data preprocessing concept	4	Lecture
8	Supervised Learning: KNN and Unsupervised Learning: K-means clustering. Assignment declaration	-	Evaluation
9	Assignment 2 evaluation + viva	1,4	Lecture
10	Logistic regression + Assignment declaration	-	Evaluation
11	Assignment 3 eval	4	Lecture
12	Project final evaluation	-	Evaluation

Appendix 1: Assessment Methods

Assessment Types	Marks
Attendance	5 %
Class Test	15 %
Assignment and Class Performance	35 %
Mid Exam	20 %
Project	25 %

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix -3: Program Outcomes

POs	Program Outcomes
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve engineering problems
PO3	An ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
PO4	An ability to design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	A knowledge of contemporary issues
PO8	An understanding of professional and ethical responsibility
PO9	An ability to function on multidisciplinary teams
PO10	An ability to communicate effectively
PO11	Project Management and Finance
PO12	A recognition of the need for, and an ability to engage in life-long learning